

A close-up photograph of a laboratory procedure. A hand wearing a bright green nitrile glove is holding a metal needle. The needle is positioned to transfer a small amount of clear liquid from a glass test tube into a well of a light blue microplate. The test tube has a metal cap and is partially filled with the same clear liquid. The background is dark and out of focus.

Aseptic technique

Aseptic Technique

A method used to prevent contamination of wounds and other susceptible sites by organism that could cause infection



In the microbiology lab we use aseptic technique to:

- Prevent contamination of the specific microorganism we are working with.
- Prevent contamination of the room and personnel with the microorganism we are working with.



Definitions:

Sterilization :

- It is a process by which an article, surface or medium is made free of all microorganisms either in vegetative or spore form.

Disinfection :

- Destruction of all pathogens or organisms capable of producing infections but not necessarily spores.
- All organisms may not be killed but the number is reduced to a level that is no longer harmful to health.

Antiseptics

- Chemical disinfectants which can safely applied to living tissues and are used to prevent infection by inhibiting the growth of microorganisms

Uses of sterilisation:

1. Sterilisation of materials, instruments used in surgical and diagnostic procedures.
2. Sterilisation of Media and reagents used in the microbiology laboratory.
3. Food and drug manufacturing to ensure safety from contaminating organisms.



Factors that influence efficacy of disinfection/sterilization

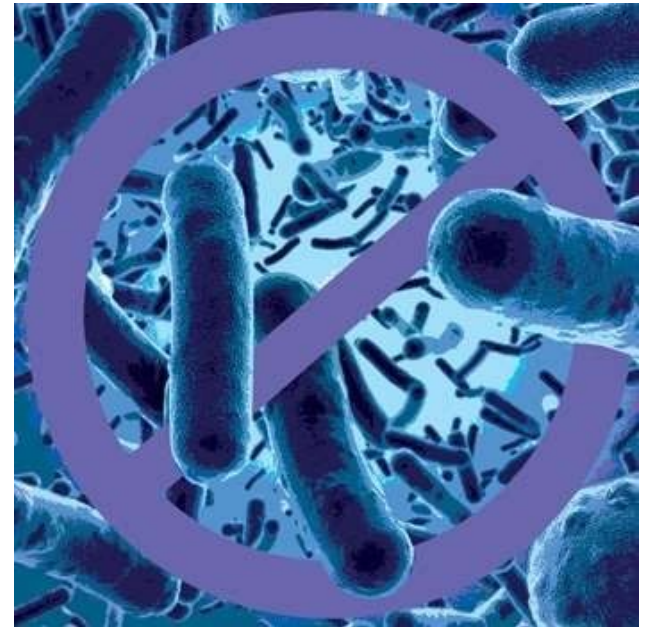
- Contact time
 - Physico-chemical environment (e.g. pH)
 - Presence of organic material
- Temperature
 - Type of microorganism
 - Number of microorganisms
- Material composition



Multiple-load capable sterilizing cabinet

How can microorganisms be killed?

- Denaturation of proteins
- Interruption of DNA synthesis/repair
- Disruption of cell membranes



Methods of sterilization

There are 2 types of sterilization: **physical and chemical:**

A-Physical methods:

1. Dry heat:

i-hot air oven → glassware

ii-flaming → inoculation loop & needle

2-moist heat(steam under high pressure):

autoclave → culture media



1- Dry heat:

i. hot air oven

- The heat takes much longer to be transferred to the organism.
- The standard setting for a hot air oven is at least 2hrs at 160 °C (320 °F).
- A rapid method heats air to 190 °C (374 °F) for 6 minutes



ii. Moist Heat

- **At 100 C°**
 - Boiling water
- **Above 100 C°**
 - Autoclaving
- **Below 100 C°**
 - Degradation of Proteins



Cycle Phases of autoclave

- Conditioning
 - Air removal
- Exposure
 - Load is heated to sterilization temp
- Exhaust
 - Bottom drain opens and steam is exhausted
- Drying
 - Dry heat revaporizes existing moisture & removes from the load



Steam Sterilization

Moist Heat

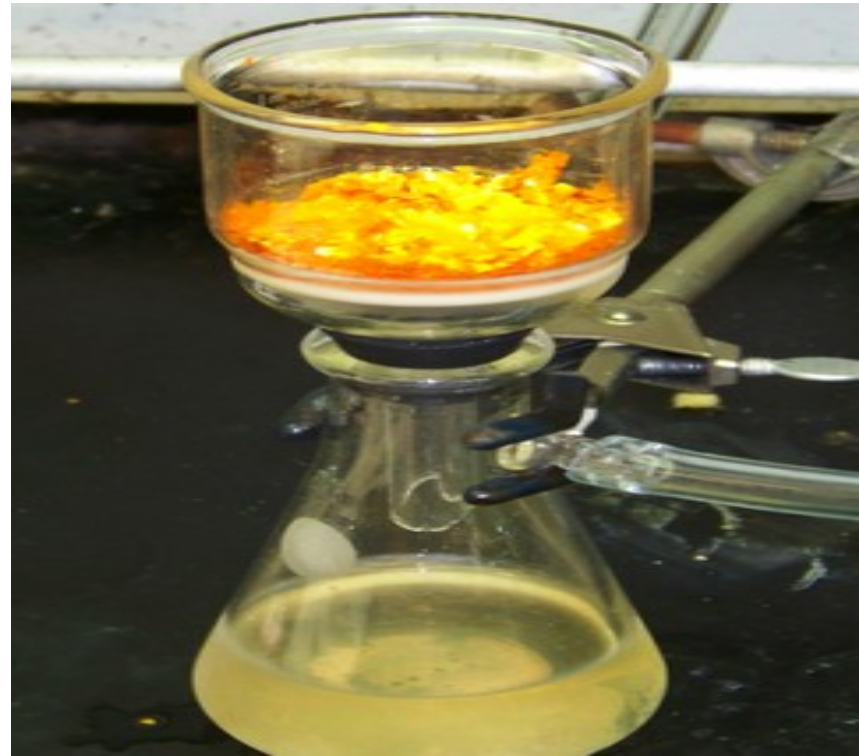
- Most economical & efficient
- Rapid processing time allows frequent availability
- Items must be able to tolerate high temps & moisture
- Works with time, temperature & moisture



Physical Methods

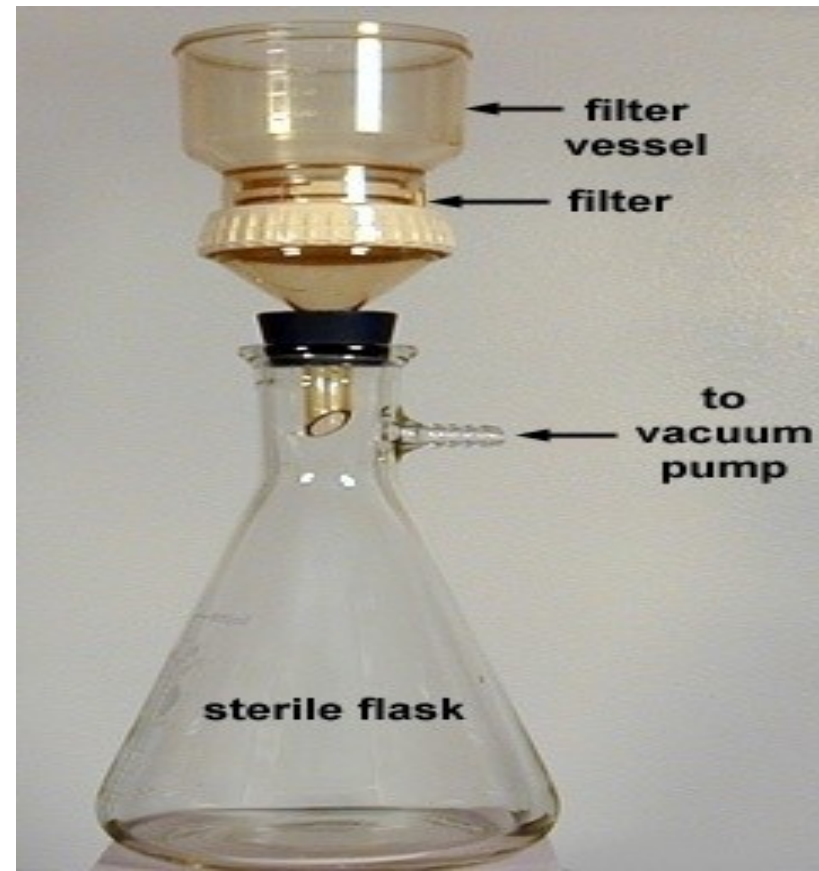
3-Filtration:

- Oversize solids in the fluid are retained.(depending on the pore size and filter thickness)
- The pore size for filtering **bacteria, yeasts, and fungi** is in the range of **0.22-0.45 μm** (filtration membranes are most popular for this purpose).



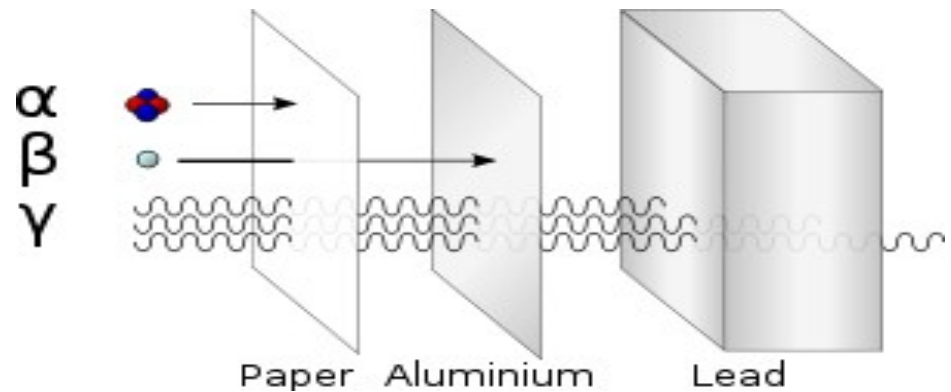
Uses of filtration

- Useful for substances which get damaged by heat.
- To sterilize sera, sugars and antibiotic solutions.
- To obtain bacteria free filtrates of clinical samples.
- Purification of water.
- Sterilize solutions
- The separation of solids from fluids



Physical Methods

- ❖ **Radiation** :is a process in which energetic particles travel through a medium or space. There are two types of *radiation*; ionizing and non-ionizing



- ❖ **X-ray**

- ❖ **Gamma radiation** are an example of ionized radiation, commercially used for sterilization of disposable items (**cold sterilization**)
- causes direct damage to DNA, produces toxic free radicals and hydrogen peroxide from water within the microbial cells

NON –IONISING RADIATION:

- ❖ **This is called hot sterilization**, Short wavelength 280 nm, UV ray inhibit DNA replication
- ❖ Infra-red radiation is used for mass sterilization of syringes. Ultraviolet radiation is used disinfecting enclosed areas like
- ❖ hospital wards
- ❖ operation rooms
- ❖ virus laboratories .



Chemical methods

It is useful for damage heat-sensitive materials such **as biological materials, fiber optic, and many plastic**

1) Gaseous sterilization:- i) Ethylene oxide

ii) Formaldehyde gas

2) Liquid sterilization:- i) Alcohol

ii) Phenol

Chemical Method

Examples Chemical sterilization includes:

- ❖ Alcohols
- ❖ Aldehydes
- ❖ Phenolics
- ❖ Oxidizing agents
- ❖ Quaternary ammonium compounds
- ❖ ethylene oxide gas
- ❖ Formaldehyde gas



Bleach

Chemical agents

Chemical agents act by

- Protein coagulation
- Disruption of the cell membrane
- Prevents normal cellular metabolism and replication
- Denature enzymes

Thank you...